

Data-Driven Deep Learning for Imaging Through Scattering Media

Architectures, Learning Paradigms, and Methodological Frontiers

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Presented at: The 6th International eSmart T.A Conference

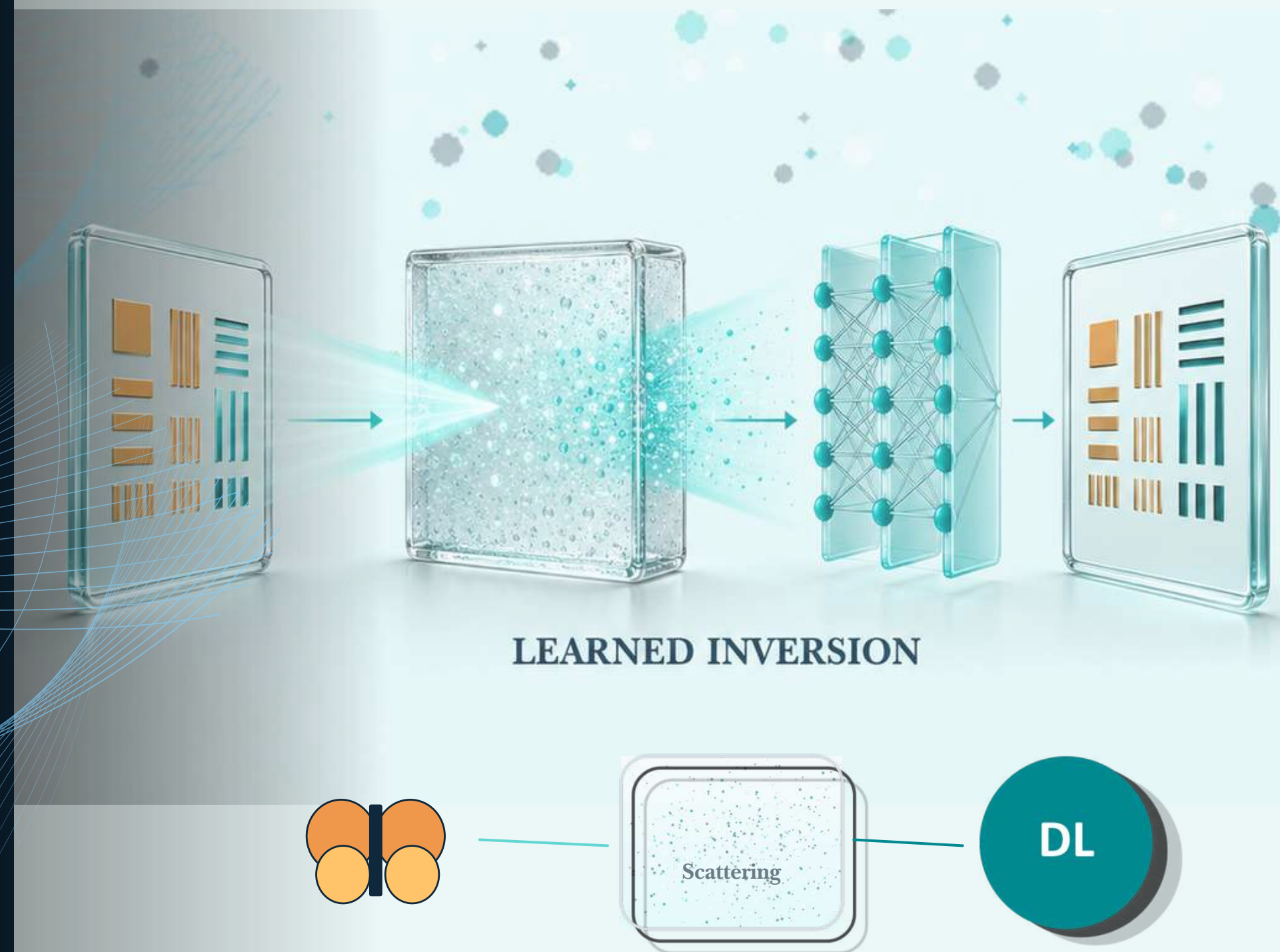
Pure data-driven reconstruction review



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Introduction

Problem context

Scattering breaks the stable correspondence between the object and the measured signal, turning image recovery into a difficult inverse problem.

Light degradation



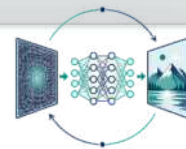
Scattering media distort light propagation.
Spatial information becomes scrambled.

Inverse challenge



Recovering hidden information is ill-posed.
Noise and medium variation reduce reliability.

Deep learning solution

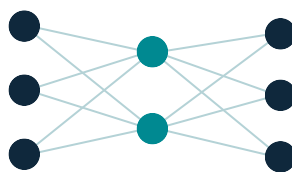


Networks learn measurement-to-object mapping.
Fast data-driven reconstruction becomes possible.

Object



Speckle



Reconstructed image

Review Scope

Pure data-driven focus

The review focuses on methods where the learned model is the primary reconstruction engine, rather than a secondary add-on to a classical pipeline.

Included studies



Deep learning-based reconstruction, restoration, or recovery.
Scattered measurements mapped directly to target outputs.
Recent peer-reviewed journal and conference papers.

Excluded directions



Classical inverse-only methods
Ghost imaging and single-pixel imaging.
Pipelines where learning is not the main reconstruction mechanism.

Speckle reconstruction

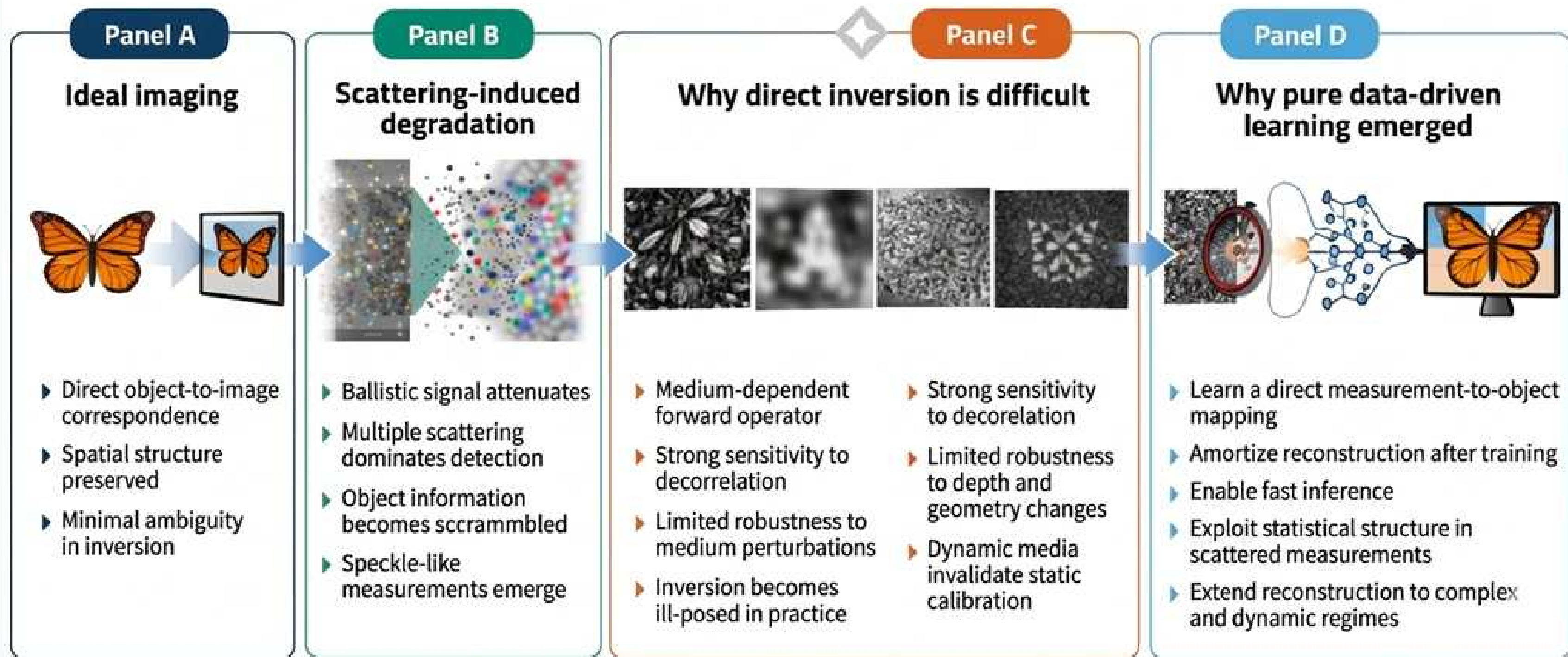
Deep learning

Inverse imaging

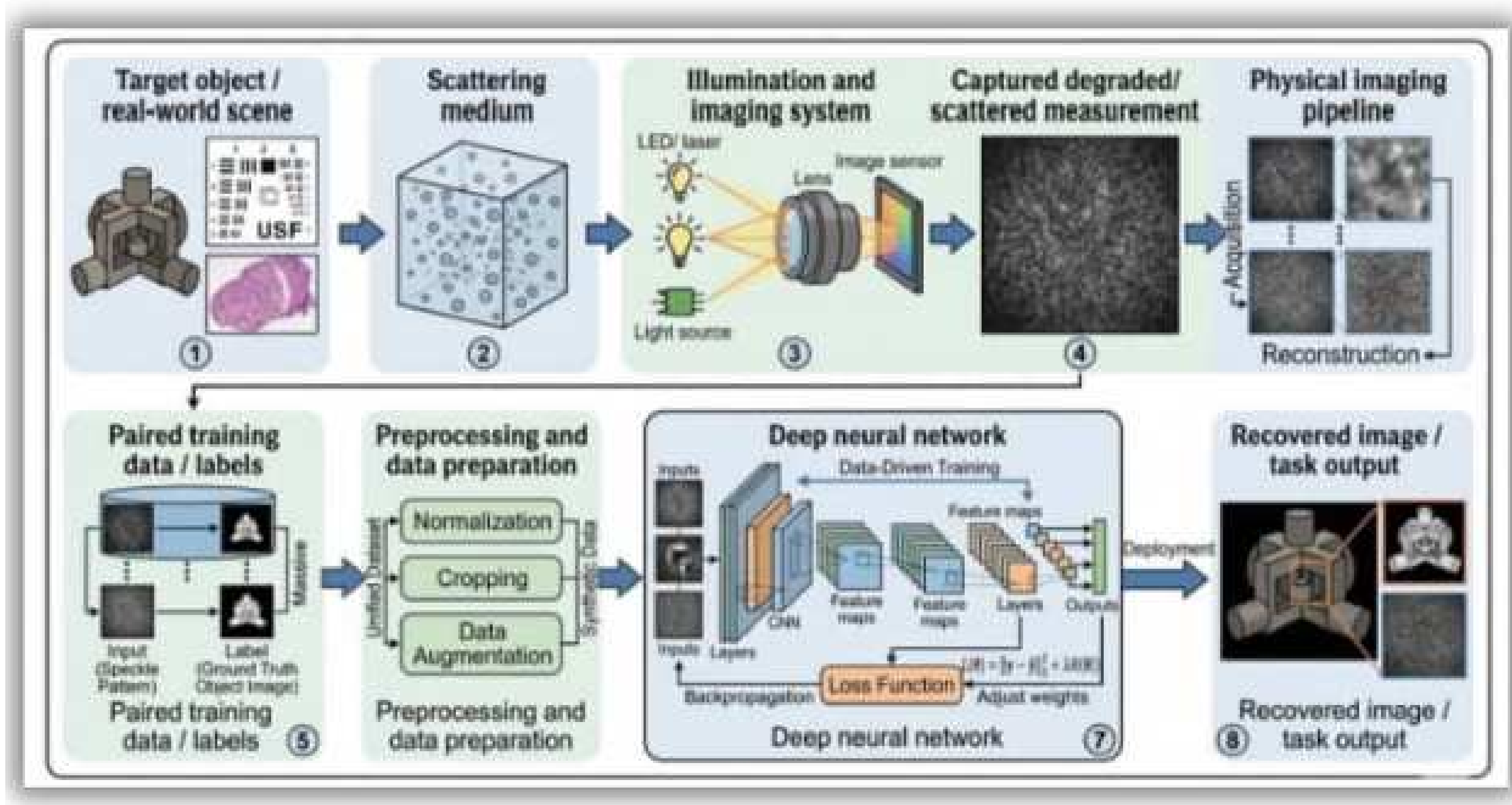
Generalization

Conceptual Transition

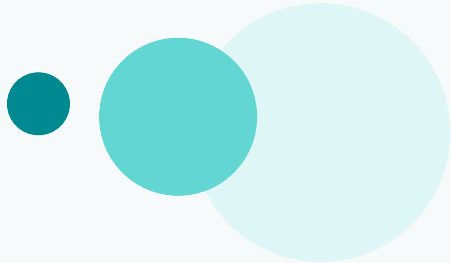
From Scattering-Induced Image Degradation to Learned Inversion: *Why Pure Data-Driven Reconstruction Emerged*



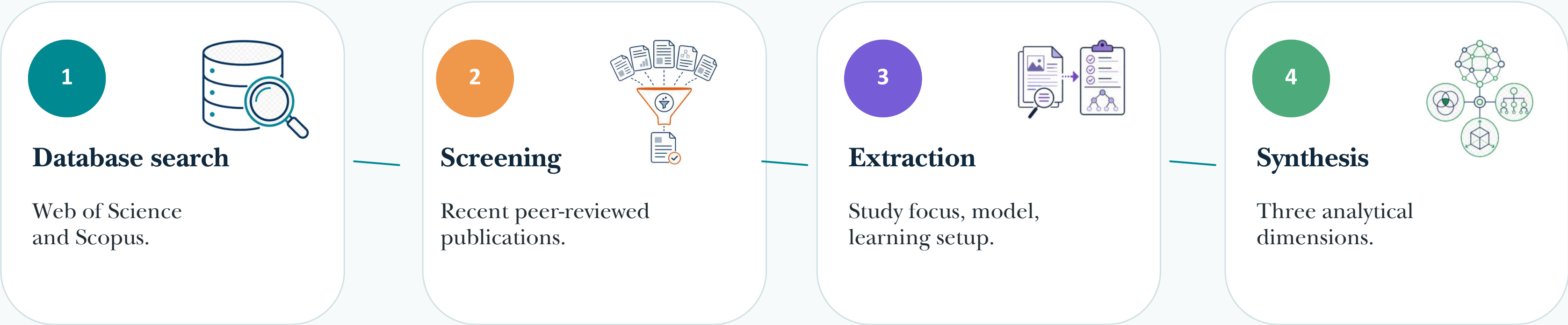
Imaging Pipeline



Methodology



Review design



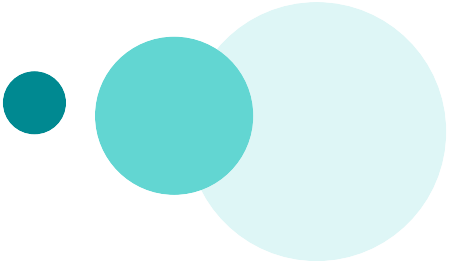
Comparative analysis dimensions:

Reconstruction Backbone

Learning Regime

System-Level Integration

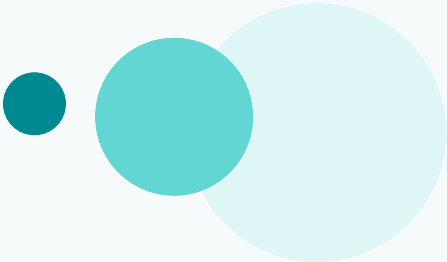
Classification of Methods



THREE-AXIS TAXONOMY OF PURE DATA-DRIVEN IMAGING THROUGH SCATTERING MEDIA

Axis	Category	Representative Studies	Role in Reconstruction	Strength	Weakness	Computational Cost	Generalization
Backbone	CNN / U-Net variants	[11], [14], [9], [18], [26], [27], [22]	Direct inverse mapping	Accurate; fast; stable; good detail recovery	Performance remains condition-sensitive under strong mismatch	Moderate	Moderate-High
	Transformer / MLP-Mixer	[17]	Global feature modeling	Strong long-range contextual interaction	Higher design and training complexity	Moderate-High	Moderate
	Hybrid CNN/Transformer	[6], [5]	Joint local-global feature modeling	Balanced local detail and global context	Reduced lightweight deployability	High	High
	Diffusion / generative restoration	[3]	Iterative restoration	Strong restoration quality; data-efficient adaptation	Slower inference due to iterative denoising	High	Moderate-High
Learning regime	Paired supervised learning	[16], [17], [18], [24], [22]	End-to-end learning from aligned input-target pairs	Strong baseline performance; stable optimization	High dependence on paired data	Moderate	Moderate
	Semi-/self-supervised learning	[8], [25], [15]	Reducing label dependence and improving robustness	Better scalability; improved adaptability	Higher methodological complexity	Moderate-High	High
	Transfer learning / fine-tuning	[9]	Adapting pretrained models to new scattering conditions	Efficient reuse of pretrained representations	Still depends on domain proximity	Moderate	Moderate-High
	Test-time / online adaptation	[3], [25]	Updating or aligning the model during deployment	Improved resilience to drift and domain shift	Additional runtime and implementation complexity	Moderate-High	High
System integration	Purely digital reconstruction	[11], [14], [3], [26]	Reconstruction performed primarily in the learned digital model	More portable across setups	Less ability to exploit acquisition-side structure	Moderate	Moderate-High
	Physics-coupled pipelines	[10], [19], [20]	Reconstruction integrated with structured acquisition or preprocessing	Strong task-specific performance	Setup-dependent; reduced portability	Moderate-High	Moderate
	Opto-electronic / hybrid optical-digital systems	[16], [24]	Partial offloading of optical reconstruction	Very low digital burden	Optical implementation constraints	Low	Moderate

Key Findings



COMPARATIVE ANALYSIS OF LEARNING PROTOCOLS, GENERALIZATION STRATEGIES, AND EVALUATION PRACTICES

Dimension	Representative Studies	Common Practice	Limitation	Impact
Learning regime	[11], [6], [10], [3], [25], [14], [9], [15], [5], [20]	• Mainly paired supervised learning • Selective use of diffusion, transfer learning, online adaptation, and self-supervision	• Strong paired-data reliance • Adaptive/self-supervised schemes increase complexity	• Learning design strongly shapes scalability and robustness
Training regime	data [11], [3], [25], [14], [9], [15], [5]	• Mostly controlled experimental paired datasets • Some studies reduce data demand via small-data training, online updating, or self-supervision	• Costly and setup-specific • Often limited in diversity across media and conditions	• Data regime remains a major bottleneck for real-world transfer
Generalization strategy	[11], [3], [26], [14], [9], [15], [5], [27]	• One-to-all training • Domain alignment and transfer learning • Residual robustness-by-design and self-supervised adaptation	• Severe mismatch and strong scattering change still degrade performance	• Generalization is increasingly treated as a core design objective
OOD / robustness evaluation	[11], [3], [25], [26], [14], [9], [15], [5], [27], [22]	• Increasing use of unseen media, displacement, depth, thickness, distance, drift, material, and angular tests	• Evidence remains fragmented • No unified robustness protocol	• Robustness evidence is growing, but remains uneven

Challenges & Future Directions

Gaps

- Paired-data and setup dependence
- Lack of unified robustness

Limitations

- Weak transfer to unseen environment
- High complexity with unclear efficiency

Challenges

- Real-world diversity and model adaptability
- Standardized quality-speed evaluation

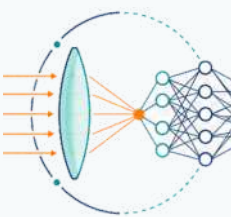
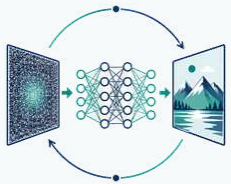
Future Directions

- Self/weak supervision and adaptive models
- Unified extreme-condition benchmarking

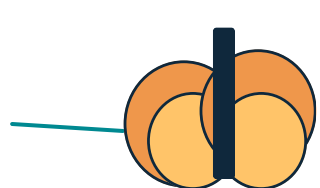
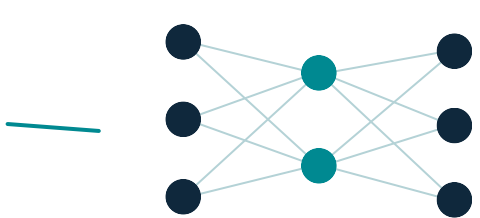
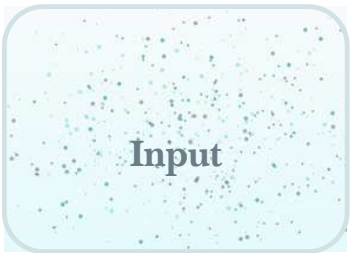
Conclusion

Main takeaways

- ✓ Deep learning has advanced imaging through scattering media.
- ✓ Methods differ by architecture, learning strategy, and integration level.
- ✓ Robustness, generalization, benchmarking, and deployment remain key challenges.
- ✓ Future systems should be accurate, adaptive, robust, and practical.



Integrated reconstruction pipeline



Physical prior

Adaptation

Benchmarking

Thank You

Questions & Discussion

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